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CHENGALPATTU DISTRICT -2026

Class : 12

Register Number | | | | | | |

COMMON QUARTERLY EXAMINATION - 2026 - 27

BIOLOGY

Time Allowed : 3.00 Hours] [Max. Marks : 70

PART - I (Bio-Botany) (Marks : 35)

SECTION - I

I. Note. (i) Answer All the questions. 8×1=8

(ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer.

1. First cell of male gametophyte in Angiosperm is
   a) Microspore b) Megaspore
   c) Nucleus d) Primary Endosperm Nucleus
2. Which of the following statement is correct?
   a) Agar is not extracted from marine algae such as seaweeds.
   b) Callus undergoes differentiation and produces somatic embryoids.
   c) Surface sterilization of explants is done by using mercuric bromide
   d) PH of the culture medium is 5.0 to 6.0
3. Plasmids are
   a) Circular protein molecules b) Required by bacteria
   c) Tiny bacteria d) Confer resistance to antibiotics
4. Which one of the following Algae is used in Single Cell Protein production?
   a) Cellulomonas b) Chlorella c) Agaricus d) Alcaligenes
5. Identify the incorrect pair.
   a) Sporopollenin - Exine of pollen grain
   b) Tapetum - Nutritive tissue for developing microspores
   c) Nucellus - Nutritive tissue for developing embryo
   d) Obturator - Directs the pollen tube into micropyle
6. haploid number in a cell is 18. The double monosomic and trisomic number will be
   a) 34 and 37 b) 34 and 35 c) 37 and 35 d) 17 and 19
7. The genes controlling the seven pea characters studied by Mendel are known to be located on how many different chromosomes?
   a) Seven b) Six c) Five d) Four
8. Who reported a lethal gene in snapdragon (Antirrhinum sp.).
   a) Carl Correns b) Hugo de Vries c) W. Bateson d) E. Baur

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SECTION - II

II. Answer Any Four Questions. 4x2=8

9. Draw and label the parts of Mature Dicot Embryo. (Capsella bursa-pastoris)
10. What is reproduction?
11. Define the term Lethal genes.
12. What is gene mapping?
13. What are the materials used to grow microorganism like Spirulina?
14. Write the differences between Somaclonal variations and Gametoclonal variations

SECTION - III

III. Answer Any Three Questions. Question 19 is Compulsory. 3x3=9

15. You are working in a biotechnology lab with a bacterium namely E.coli. How will you cut the nucleotide sequence? explain it.
16. Enumerate the characteristic features of Entomophilous flowers.
17. Write the differences between linkage and crossing over
18. What are the advantages of Artificial seeds.
19. How will you explain incomplete dominance at the molecular level?

SECTION - IV

IV. Answer The Following Questions In Detail. 2x5=10

20. a) Describe dominant epistasis with an example.

(OR)

b) With a suitable diagram explain the structure of an ovule.

21. a) Explain the steps involved in Recombinant DNA Technology

(OR)

b) (i) Explain the basic concepts involved in plant tissue culture.

(ii) Name the chemicals used in gene transfer.

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PART – II (BIO-ZOOLOGY) (Max. Marks: 35)

SECTION – I

I. Choose the correct appropriate answer. 8x1=8

1. Colostrum is rich in

a) Ig E b) Ig A c) Ig D d) Ig M

2. A procedure that finds use in testing for genetic disorders but also misused for foeticide is ----------

a) Vasectomy b) Tubectomy c) Aminocentesis d) Artificial insemination

3. B Cells that produce and release large amounts of antibody are called ----------

a) Basophils b) Plasma cells c) Memory cells d) killer cells

4. A mRNA molecule is produced by

a) Replication b) Transcription c) Duplication d) Translation

5. Which of the following is used as 'Clot buster'

a) Ethanol b) Statins c) Cyclosporin - A d) Streptokinase

6. The mosquito borne disease in humans causing chronic inflammation of lymphatic vessel is

a) elephantiasis b) ascariasis c) ring worm d) amoebiasis

7. ZW - ZZ system of sex determination occurs in

a) Fishes b) Reptiles c) Birds d) All the above

8. **Assertion** : The number of white winged moth decreased after industrialisation in England.

Reason : Effects of industrialisation were marked in rural area of England.

a) A and R are correct, R is correct explanation of A

b) A and R are correct, R is not correct explanation of A

c) A is correct and R is false

d) A and R are false.

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SECTION - II

II. Answer any Four questions. 4x2=8

9 List any 3 household products along with names of micro - organisms producing them

10. **Give reasons:** A male honey bee has 16 Chromosomes whereas female has 32 Chromosome

11. **Differentiate :** Leading and Lagging Strand.

12. How will you diagnose for AIDS?

13. Give 4 examples for Vestigeal organs.

14. **Correct the Statements.**

a) Multiload 375 is a hormone releasing IUD.

b) Transfer of an ovum collected from donor into the fallopian tube is called ZIFT

SECTION - III

III. Answer any Three of the following questions. Q.NO: 19 is Compulsory. 3x3=9

15. Write a short note on Ideonella sakaiensis.

16. Tabulate the causative agent and symptoms of one bacterial and one viral sexually transmitted

17. How is polyspermy avoided?

18. What are multiple alleles? Give example.

19. During medical investigation, an infant was found to posses an extra copy of chromosome 21 Identify the disorder. List the symptoms the child is likely to develop.

SECTION - IV

IV. Answer in Detail. 2x5=10

20. a) Explain the structure of immunoglobulin with a neat labelled diagram.

(OR)

b) Mention the main objections to Darwinism.

21. a) Write down the salient features of genetic code.

(OR)

b) Explain in detail about the extra embryonic membranes seen in human.

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